

Capstone Project Proposal (Data Analysis Pathway)

Project Title: *Keystone Data Analysis on AI Adoption and Workforce Impact (2023-2026)*

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GitHub Repository: [HemDesh/Data Analysis on AI Adoption and Workforce Impact: Data Analysis on AI Adoption and impact on Workforce globally](https://github.com/HemDesh/Data-Analysis-on-AI-Adoption-and-Workforce-Impact)

Project Overview

The overall purpose of the project.

This project analyzes global AI adoption trends across industries, regions, and companies using exploratory data analysis (EDA) techniques. The project analyzes the data and presents how AI investment, adoption maturity, automation, and reskilling efforts impact productivity, revenue growth, and employment.

What am I trying to analyze, explain, or uncover

The project includes:

Data cleaning and preprocessing

Handling missing and duplicate values

Statistical analysis

Correlation analysis

Multivariate analysis

charts and heatmaps

Business insights extraction

Why does this matter?

AI is a big in tech industries. Not only tech industries, AI is changing every aspect of human lives. Any new technologies have advantages & disadvantages. Similarly, AI also have advantages and disadvantages in productivity, revenue growth, and employment, which matters to everyone.

Data Sources

List the datasets you will use. For each, include:

- **Dataset Source** <https://www.kaggle.com/datasets/mohankrishnathalla/global-ai-adoption-and-workforce-impact-dataset>

- Dataset 1 - AI Company Adoption Data
 - This dataset is a company wise collection of AI adoption data for past a few years across various countries.
 - **Relevant fields** I plan to use
 - response_id
 - company_id
 - survey_year
 - quarter
 - country
 - industry
 - company_size
 - num_employees
 - annual_revenue_usd_millions
 - company_founding_year
 - company_age
 - company_age_group
 - ai_adoption_rate
 - ai_adoption_stage
 - years_using_ai
 - ai_primary_tool
 - num_ai_tools_used
 - ai_use_case
 - ai_projects_active
 - ai_training_hours
 - ai_budget_percentage
 - ai_maturity_score
 - ai_failure_rate
 - ai_investment_per_employee
 - regulatory_compliance_score
 - data_privacy_level
 - ai_ethics_committee
 - ai_risk_management_score
 - remote_work_percentage
 - employee_satisfaction_score
 - task_automation_rate
 - time_saved_per_week
 - productivity_change_percent
 - jobs_displaced
 - jobs_created
 - reskilled_employees
 - revenue_growth_percent
 - cost_reduction_percent
 - innovation_score
 - customer_satisfaction
 - survey_source
 - data_collection_method

- Dataset 2 - Country Wise Technology Data
 - This dataset gives country wise some key information
 - **Relevant fields** I plan to use
 - country
 - region
 - gdp_per_capita
 - internet_penetration
 - digital_maturity_index
 - country_ai_policy
 - ai_patent_filings_2024
 - ai_researchers_per_million

Research Objectives

Main questions and objectives.

- **Primary Question(s):** the key relationships or patterns I want to test.
 - Does the AI adoption, job displacement, transforming company's workforce have any influence of the country's GDP
 - How the internet penetration can affect AI progress
- **Secondary/Exploratory Questions:** additional comparisons or trends I want to explore.
 - Also find out the effect of digital maturity & country's AI policy

Why these objectives are valuable or interesting.

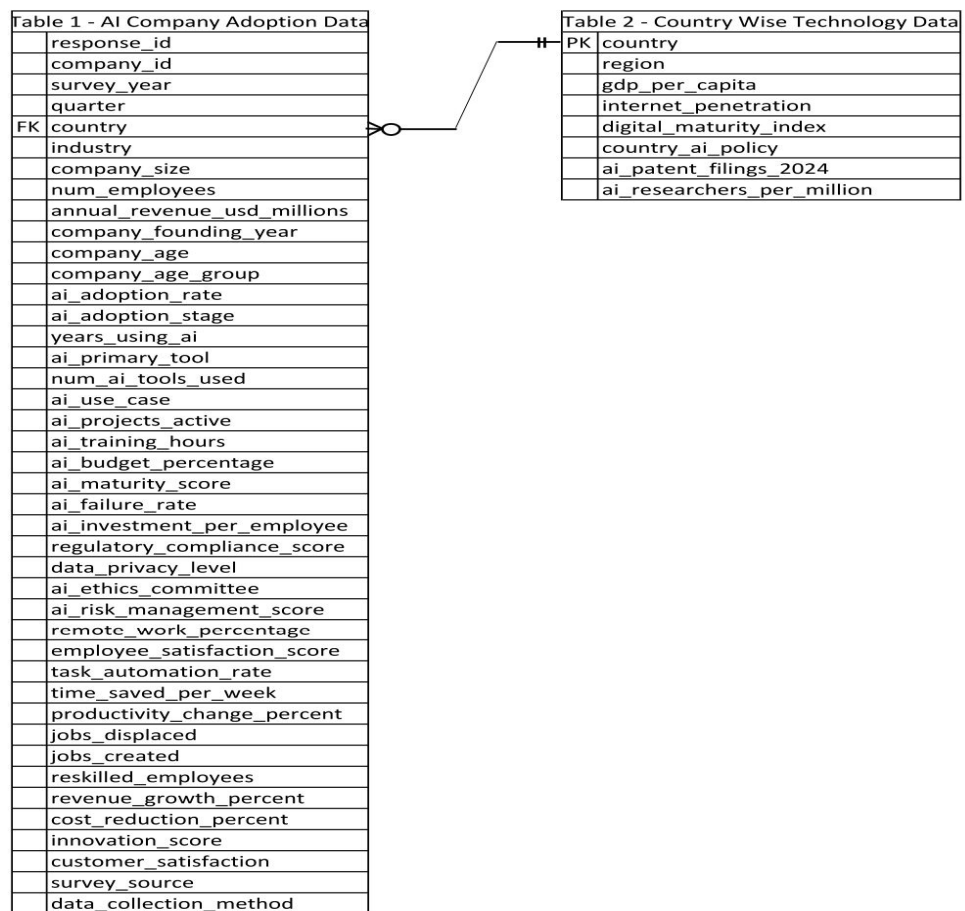
- During the project work in the last module, I already have used the dataset 1 to analyze the data
- I would like to extend that analysis by country and find out some key answers like

Data Preparation Approach

Data preparation for analysis.

- How you will align or merge datasets (keys, years, geography).
 - I have prepared ER diagram and that shows Dataset 1 and Dataset 2 are connected by a field "country"
- How you will address missing values (replacement, dropping rows, etc.).
 - There are no missing values in the dataset
- How you will address outliers or extreme values.
- New variables or indices I may create.

- AI adoption by country
- job displacement index by country
- AI adoption by digital maturity
- AI adoption by AI policy
- How you will structure your relational database (at minimum two tables from different sources, joined through a common key). Here is the ER diagram



Current Status

Summarize what you have already completed. Examples:

- Data acquired and inspected - complete
- Early cleaning or standardization steps - complete

- Prototype functions or exploratory plots - complete
- The country codes in the both the datasets are consistent and can be successfully merged.

Deliverables (Remaining Work)

List the specific things you still need to complete for your project. Break them into **required** tasks (must be done to meet project requirements) and **stretch** tasks (optional, if time allows).

- **Required Tasks (must be completed):**

I still have to build SQLite schema and load tables; and produce 3 visualizations; write README and data dictionary.)

Project Timeline

Outline the sequence of major steps or milestones leading up to the deadline. You may structure this as weeks, phases, or checkpoints. Examples:

- Phase 1: Acquire and clean data, set up database in sqlite 07/04/2026.
- Phase 2: Exploratory analysis, build functions, first draft visuals – 07/11/2026.
- Phase 3: Deeper analysis, refine visuals, draft report – 07/18/2026.
- Phase 4: Finalize deliverables, polish repo, record presentation – 07/25/2026.